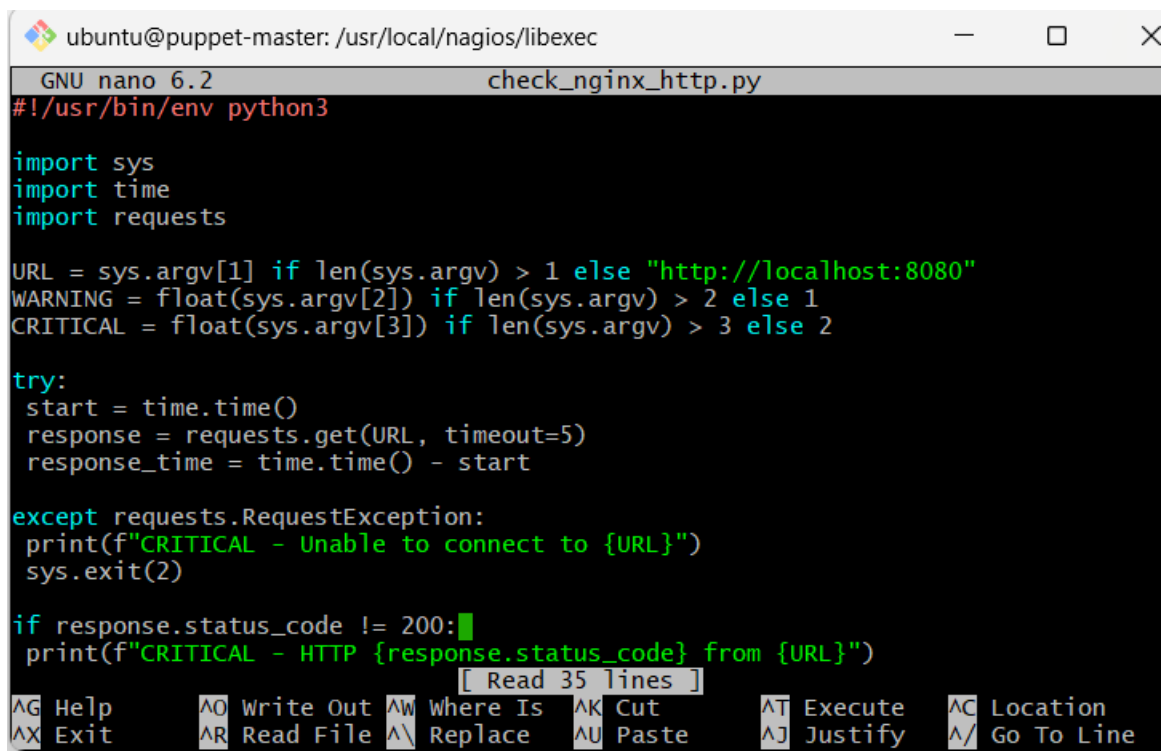


Nagios Mastery Part – 2

4. Production-Grade Custom Python HTTP Monitoring Plugin with requests, Response-Time Thresholds and Nagios Integration



```
ubuntu@puppet-master: /usr/local/nagios/libexec
GNU nano 6.2 check_nginx_http.py
#!/usr/bin/env python3

import sys
import time
import requests

URL = sys.argv[1] if len(sys.argv) > 1 else "http://localhost:8080"
WARNING = float(sys.argv[2]) if len(sys.argv) > 2 else 1
CRITICAL = float(sys.argv[3]) if len(sys.argv) > 3 else 2

try:
    start = time.time()
    response = requests.get(URL, timeout=5)
    response_time = time.time() - start

except requests.RequestException:
    print(f"CRITICAL - Unable to connect to {URL}")
    sys.exit(2)

if response.status_code != 200:
    print(f"CRITICAL - HTTP {response.status_code} from {URL}")

[ Read 35 lines ]

^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

```
ubuntu@puppet-master: /usr/local/nagios/libexec
GNU nano 6.2 check_nginx_http.py *
except requests.RequestException:
    print(f"CRITICAL - Unable to connect to {URL}")
    sys.exit(2)

if response.status_code != 200:
    print(f"CRITICAL - HTTP {response.status_code} from {URL}")
    sys.exit(2)

if response_time >= CRITICAL:
    print(f"CRITICAL - HTTP 200,response time {response_time:.3f}s")
    sys.exit(2)

elif response_time >= WARNING:
    print(f"WARNING - HTTP 200,response time {response_time:.3f}s")
    sys.exit(1)

else:
    print(f"OK - HTTP 200,response time {response_time:.3f}s")
    sys.exit(0)

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
elif response_time >= WARNING:
    print(f"WARNING - HTTP 200,response time {response_time:.3f}s")
    sys.exit(1)

else:
    print(f"OK - HTTP 200,response time {response_time:.3f}s")
    sys.exit(0)

ubuntu@puppet-master:/usr/local/nagios/libexec$ ./check_nginx_http.py http://localhost:8080 1 2
OK - HTTP 200,response time 0.004s
ubuntu@puppet-master:/usr/local/nagios/libexec$
```

← → ↻ Not secure 15.206.174.155/nagios/

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer All Bookmarks

Nagios

General

- Home
- Documentation
- Current Status**
- Tactical Overview
- Map
- Hosts
- Services
- Host Groups
- Summary
- Grid
- Service Groups
- Summary
- Grid
- Problems
- Services (Unhandled)
- Hosts (Unhandled)
- Network Outages
- Quick Search:

Reports

- Availability
- Trends
- Alerts
- History
- Summary
- Histogram
- Notifications
- Event Log
- System**
- Comments
- Downtime
- Process Info
- Performance Info

Service Information

Last Updated: Wed Sep 2 13:08:20 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service
NGINX HTTP Response
On Host
localhost
(localhost)

Member of
No servicegroups.

127.0.0.1

Service State Information

Current Status: **OK** (for 0d 0h 0m 45s+)
Status Information: HTTP 200,response_time 0.000370s - Endpoint OK
Performance Data: 1/3 (HARD state)
Current Attempt: 09-02-2026 13:07:58
Last Check Time: 09-02-2026 13:07:58
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.023 seconds
Next Scheduled Check: 09-02-2026 13:08:58
Last State Change: N/A
Last Notification: N/A (notification 0)
Is This Service Flapping? **NO** (0.00% state change)
In Scheduled Downtime? **NO**
Last Update: 09-02-2026 13:08:15 (0d 0h 0m 5s ago)

Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

- ✗ Disable active checks of this service
- ⌚ Re-schedule the next check of this service
- ❓ Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✗ Stop obsessing over this service
- ✗ Disable notifications for this service
- 🔊 Send custom service notification
- 🕒 Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments

🔍 Add a new comment 🗑 Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Page Tour

```

ubuntu@puppet-master: ~
Checking global event handlers...
Checking obsessive compulsive processor commands...
Checking misc settings...

Total Warnings: 0
Total Errors: 0

Things look okay - No serious problems were detected during the pre-flight check
ubuntu@puppet-master:~$ sudo systemctl restart nagios
ubuntu@puppet-master:~$ sudo systemctl stop nagios
ubuntu@puppet-master:~$ sudo systemctl status nagios
o nagios.service - Nagios Core 4.5.10
   Loaded: loaded (/lib/systemd/system/nagios.service; enabled; vendor preset>
   Active: inactive (dead) since Wed 2026-09-02 13:09:26 UTC; 7s ago
     Docs: https://www.nagios.org/documentation
   Process: 1725 ExecStartPre=/usr/local/nagios/bin/nagios -v /usr/local/nagio>
   Process: 1726 ExecStart=/usr/local/nagios/bin/nagios -d /usr/local/nagios/e>
   Process: 1828 ExecStop=/usr/bin/kill -s TERM ${MAINPID} (code=exited, statu>
   Process: 1829 ExecStopPost=/usr/bin/rm -f /usr/local/nagios/var/rw/nagios.c>
   Main PID: 1727 (code=exited, status=0/SUCCESS)
      CPU: 133ms

Sep 02 13:07:35 puppet-master nagios[1727]: wproc: Registry request: name=Core >
Sep 02 13:07:35 puppet-master nagios[1727]: wproc: Registry request: name=Core >
Sep 02 13:07:35 puppet-master nagios[1727]: Successfully launched command file >

```

Nagios: 15.206.174.155

15.206.174.155/nagios/

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer

All Bookmarks

Nagios

General

HomeDocumentationCurrent StatusTactical OverviewMapHostsServicesHost GroupsSummaryGridService GroupsSummaryGridProblemsServices (Unhandled)Hosts (Unhandled)Network OutagesQuick Search:ReportsAvailabilityTrendsAlertsHistorySummaryHistogramNotificationsEvent LogSystemCommentsDowntimeProcess InfoPerformance Info

Service Information

Last Updated: Wed Sep 2 13:12:08 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

NGINX HTTP Response

On Hostlocalhost(localhost)

Member ofNo servicegroups.

127.0.0.1

Service State Information

Current Status:CRITICAL (for 0d 0h 0m 19s)
Status Information:Unable to connect to http://localhost:8080
Performance Data:
Current Attempt:1/3 (SOFT state)
Last Check Time:09-02-2026 13:11:49
Check Type:ACTIVE
Check Latency / Duration:0.001 / 0.010 seconds
Next Scheduled Check:09-02-2026 13:12:49
Last State Change:09-02-2026 13:11:49
Last Notification:N/A (notification 0)
Is This Service Flapping?NO (0.00% state change)
In Scheduled Downtime?NO
Last Update:09-02-2026 13:12:00 (0d 0h 0m 8s ago)
Active Checks:ENABLED
Passive Checks:ENABLED
Obsessing:ENABLED
Notifications:ENABLED
Event Handler:ENABLED
Flap Detection:ENABLED

Service Commands

Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Disable notifications for this service
Delay next service notification
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service

Service Comments

Add a new commentDelete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

Page Tour

Nagios

General

HomeDocumentationCurrent StatusTactical OverviewMapHostsServicesHost GroupsSummaryGridService GroupsSummaryGridProblemsServices (Unhandled)Hosts (Unhandled)Network OutagesQuick Search:ReportsAvailabilityTrendsAlertsHistorySummaryHistogramNotificationsEvent LogSystemCommentsDowntimeProcess InfoPerformance Info

Service Information

Last Updated: Wed Sep 2 13:14:01 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

NGINX HTTP Response

On Hostlocalhost(localhost)

Member ofNo servicegroups.

127.0.0.1

Service State Information

Current Status:CRITICAL (for 0d 0h 0m 12s)
Status Information:Unable to connect to http://localhost:8080
Performance Data:
Current Attempt:3/3 (HARD state)
Last Check Time:09-02-2026 13:13:49
Check Type:ACTIVE
Check Latency / Duration:0.000 / 0.010 seconds
Next Scheduled Check:09-02-2026 13:14:49
Last State Change:09-02-2026 13:13:49
Last Notification:09-02-2026 13:13:49 (notification 1)
Is This Service Flapping?NO (6.25% state change)
In Scheduled Downtime?NO
Last Update:09-02-2026 13:14:00 (0d 0h 0m 1s ago)
Active Checks:ENABLED
Passive Checks:ENABLED
Obsessing:ENABLED
Notifications:ENABLED
Event Handler:ENABLED
Flap Detection:ENABLED

Service Commands

Disable active checks of this service
Re-schedule the next check of this service
Submit passive check result for this service
Stop accepting passive checks for this service
Stop obsessing over this service
Acknowledge this service problem
Disable notifications for this service
Delay next service notification
Send custom service notification
Schedule downtime for this service
Disable event handler for this service
Disable flap detection for this service
Clear flapping state for this service

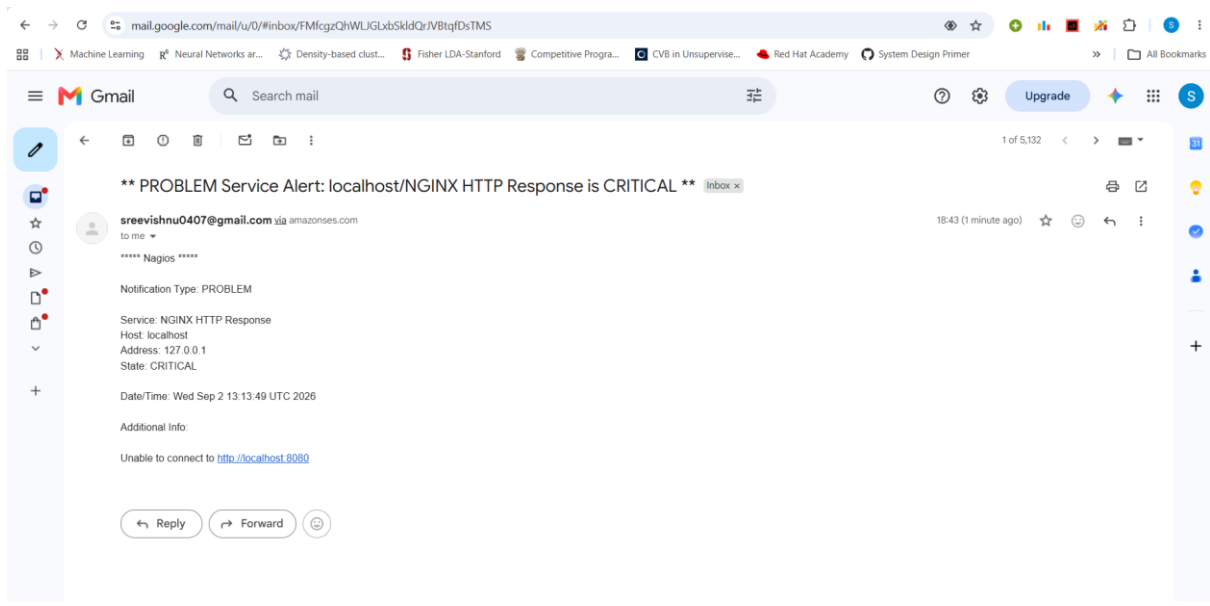
Service Comments

Add a new commentDelete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

Page Tour



5. Production-Style Custom Bash Disk Usage Monitoring Plugin with Dynamic Filesystem Thresholds and Nagios Integration

```
ubuntu@ip-172-31-2-105: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_disk_usage  
#!/bin/bash  
  
FILESYSTEM="${1:-/}"  
WARNING="${2:-80}"  
CRITICAL="${3:-90}"  
  
USAGE=$(df -P "$FILESYSTEM" | awk 'NR==2 {print $5}' | tr -d '%')  
  
if [ -z "$USAGE" ];then  
    echo "UNKNOWN - Unable to determine disk usage for $FILESYSTEM"  
    exit 3  
fi  
  
if [ "$USAGE" -ge "$CRITICAL" ];then  
    echo "CRITICAL - $FILESYSTEM usage is ${USAGE}%"  
    exit 2  
  
elif [ "$USAGE" -ge "$WARNING" ];then  
    echo "WARNING - $warning usage is ${USAGE}%"  
    exit 1  
  
[ Read 25 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@ip-172-31-2-105: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_disk_usage
USAGE=$(df -P "$FILESYSTEM" | awk 'NR==2 {print $5}' | tr -d '%')

if [ -z "$USAGE" ];then
    echo "UNKNOWN - Unable to determine disk usage for $FILESYSTEM"
    exit 3
fi

if [ "$USAGE" -ge "$CRITICAL" ];then
    echo "CRITICAL - $FILESYSTEM usage is ${USAGE}%"
    exit 2

elif [ "$USAGE" -ge "$WARNING" ];then
    echo "WARNING - $warning usage is ${USAGE}%"
    exit 1

else
    echo "OK - $FILESYSTEM usage is ${USAGE}%"
    exit 0
fi

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@ip-172-31-2-105: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg

define command {
    command_name    check_local_disk
    command_line    $USER1$/check_disk -w $ARG1$ -c $ARG2$ -p $ARG3$
}

define command {
    command_name    check_disk_usage
    command_line    $USER1$/check_disk_usage $ARG1$ $ARG2$ $ARG3$
}

define command {
    command_name    check_local_load
    command_line    $USER1$/check_load -w $ARG1$ -c $ARG2$
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@ip-172-31-2-105: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/localhost.cfg

define service {
    use                local-service          ; Name of service template
    host_name          localhost
    service_description Root Partition
    check_command       check_local_disk!20%!10!/>
}

define service {
    use local-service
    host_name localhost
    service_description Root Disk Usage
    check_command check_disk_usage!/!80!90
    check_interval 1
    retry_interval 1
    max_check_attempts 3
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@ip-172-31-2-105: ~
ubuntu@ip-172-31-2-105:~$ sudo nano /usr/local/nagios/etc/objects/commands.cfg
ubuntu@ip-172-31-2-105:~$ sudo nano /usr/local/nagios/etc/objects/localhost.cfg
ubuntu@ip-172-31-2-105:~$ sudo /usr/local/nagios/bin/nagios -v /usr/local/nagios
/etc/nagios.cfg

Nagios Core 4.5.10
Copyright (c) 2009-present Nagios Core Development Team and Community Contributors
Copyright (c) 1999-2009 Ethan Galstad
Last Modified: 2025-10-28
License: GPL

Website: https://www.nagios.org
Reading configuration data...
    Read main config file okay...
    Read object config files okay...

Running pre-flight check on configuration data...

Checking objects...
    Checked 9 services.
    Checked 1 hosts.
    Checked 1 host groups.
    Checked 0 service groups.
```

```
ubuntu@ip-172-31-2-105: ~  
Checked 9 services.  
Checked 1 hosts.  
Checked 1 host groups.  
Checked 0 service groups.  
Checked 1 contacts.  
Checked 1 contact groups.  
Checked 25 commands.  
Checked 5 time periods.  
Checked 0 host escalations.  
Checked 0 service escalations.  
Checking for circular paths...  
Checked 1 hosts  
Checked 0 service dependencies  
Checked 0 host dependencies  
Checked 5 timeperiods  
Checking global event handlers...  
Checking obsessive compulsive processor commands...  
Checking misc settings...  
  
Total Warnings: 0  
Total Errors: 0  
  
Things look okay - No serious problems were detected during the pre-flight check  
ubuntu@ip-172-31-2-105:~$
```

Nagios 3.111.36.114

← → ↻ 🔒 Not secure 3.111.36.114/nagios/

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer

Nagios

General
Home
Documentation
Current Status
Tactical Overview
Map
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends
Alerts
History
Summary
Histogram
Notifications
Event Log
System
Comments
Downtime
Process Info
Performance Info

Service Information

Last Updated: Sat Sep 5 06:17:56 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

Root Disk Usage

On Host
localhost
(localhost)

Member of
No servicegroups.

1270.01

Service State Information

Current Status: **OK** (for 0d 0h 9m 21s)
Status Information: OK - / usage is 31%
Performance Data: 1/3 (HARD state)
Current Attempt: 09-05-2026 06:17:35
Last Check Time: 09-05-2026 06:17:35
Check Type: ACTIVE
Check Latency / Duration: 0.000 / 0.007 seconds
Next Scheduled Check: 09-05-2026 06:18:35
Last State Change: 09-05-2026 06:08:35
Last Notification: 09-05-2026 06:08:35 (notification 0)
Is This Service Flapping? **NO** (10.00% state change)
In Scheduled Downtime? **NO**
Last Update: 09-05-2026 06:17:55 (0d 0h 0m 1s ago)
Active Checks: **ENABLED**
Passive Checks: **ENABLED**
Obsessing: **ENABLED**
Notifications: **ENABLED**
Event Handler: **ENABLED**
Flap Detection: **ENABLED**

Service Commands

✗ Disable active checks of this service
🕒 Re-schedule the next check of this service
? Submit passive check result for this service
✗ Stop accepting passive checks for this service
✗ Stop obsessing over this service
✗ Disable notifications for this service
📧 Send custom service notification
🕒 Schedule downtime for this service
✗ Disable event handler for this service
✗ Disable flap detection for this service
✗ Clear flapping state for this service

Service Comments

🗨 Add a new comment 🗑 Delete all comments

Entry	Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it								

Page Tour

Let's trigger Criticality:

```
define service {
    use local-service
    host_name localhost
    service_description Root Disk Usage
    check_command check_disk_usage!/!20!30
    check_interval 1
    retry_interval 1
    max_check_attempts 3
}
```

The screenshot displays the Nagios web interface in a browser. The address bar shows 'Nagios: 3.111.36.114' and the URL '3.111.36.114/nagios/'. The left sidebar contains a navigation menu with categories: General, Reports, and System. The main content area is titled 'Service Information' and shows details for the service 'Root Disk Usage' on host 'localhost'. The service is in a 'CRITICAL' state, with a message '(for 0d 0h 0m 17s)'. The 'Service State Information' section provides a detailed overview of the service's current status, performance data, and configuration. The 'Service Commands' section lists various actions that can be performed on the service. The 'Service Comments' section at the bottom shows that there are no comments associated with this service.

Service Information
Last Updated: Sat Sep 5 06:55:52 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

Service
Root Disk Usage
On Host **localhost**
(localhost)
Member of **No servicegroups.**
127.0.0.1

Service State Information

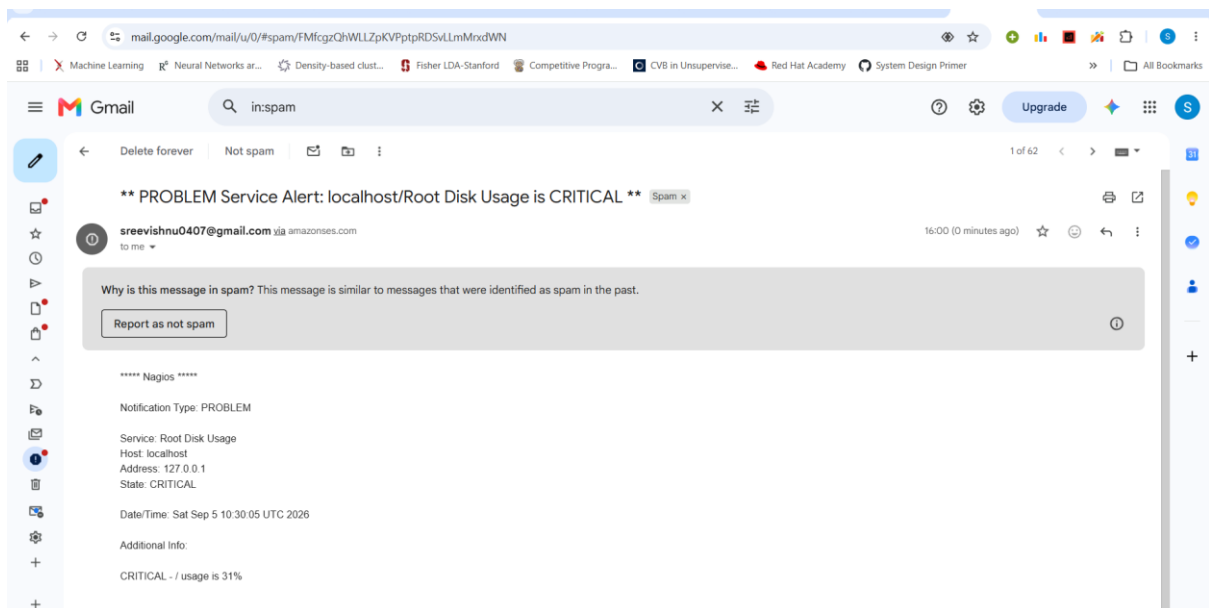
Current Status:	CRITICAL (for 0d 0h 0m 17s)
Status Information:	CRITICAL - / usage is 32%
Performance Data:	
Current Attempt:	1/3 (SOFT state)
Last Check Time:	09-05-2026 06:55:35
Check Type:	ACTIVE
Check Latency / Duration:	0:001 / 0:010 seconds
Next Scheduled Check:	09-05-2026 06:56:35
Last State Change:	09-05-2026 06:55:35
Last Notification:	N/A (notification 0)
Is This Service Flapping?	NO (0.00% state change)
In Scheduled Downtime?	NO
Last Update:	09-05-2026 06:55:42 (0d 0h 0m 10s ago)
Active Checks:	ENABLED
Passive Checks:	ENABLED
Obsessing:	ENABLED
Notifications:	ENABLED
Event Handler:	ENABLED
Flap Detection:	ENABLED

Service Commands

- ✗ Disable active checks of this service
- ⌚ Re-schedule the next check of this service
- ❓ Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✗ Stop obsessing over this service
- ✗ Disable notifications for this service
- ⌚ Delay next service notification
- 📧 Send custom service notification
- ⌚ Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments
Add a new comment Delete all comments
Entry Time Author Comment Comment ID Persistent Type Expires Actions
This service has no comments associated with it

Page Tour



Quick revamp of all steps for how mail was configured:

- As u know it's always said that the more u practice and understand, the better it gets

PART A — Configure Postfix → Amazon SES

1. Install packages

`sudo apt update`

`sudo apt install -y postfix mailutils libsasl2-modules`

Purpose:

Postfix → SMTP mail server

mailutils → gives you the `mail` command for testing

libsasl2-modules → SASL authentication support

2. Configure SES as the relay

```
sudo postconf -e "relayhost = [email-smtp.ap-south-1.amazonaws.com]:587"
```

This tells Postfix:

Don't deliver mail directly. Send outgoing mail through Amazon SES SMTP on port 587.

Then:

```
sudo postconf -e "smtp_sasl_auth_enable = yes"
```

```
sudo postconf -e "smtp_sasl_security_options = noanonymous"
```

```
sudo postconf -e "smtp_sasl_password_maps = hash:/etc/postfix/sasl_passwd"
```

```
sudo postconf -e "smtp_use_tls = yes"
```

```
sudo postconf -e "smtp_tls_security_level = encrypt"
```

```
sudo postconf -e "smtp_tls_note_starttls_offer = yes"
```

```
sudo postconf -e "smtp_tls_CAfile = /etc/ssl/certs/ca-certificates.crt"
```

3. Create credentials

```
sudo nano /etc/postfix/sasl_passwd
```

Put:

```
[email-smtp.ap-south-1.amazonaws.com]:587:SMTP_USERNAME:SMTP_PASSWORD
```

Then:

```
sudo chmod 600 /etc/postfix/sasl_passwd
```

```
sudo postmap /etc/postfix/sasl_passwd
```

Think:

```
sasl_passwd
```

↓

human-readable credentials

↓ postmap

sasl_passwd.db

↓

Postfix lookup database

4. Restart:

```
sudo systemctl restart postfix
```

```
sudo systemctl status postfix --no-pager
```

5. Test Postfix independently

Before involving Nagios, prove that Postfix can send:

```
echo "SES SMTP test" | mail \  
-r "YOUR_VERIFIED_EMAIL" \  
-s "SES Test" \  
"YOUR_VERIFIED_EMAIL"
```

Then:

```
sudo tail -30 /var/log/mail.log
```

If debugging:

```
sudo tail -f /var/log/mail.log
```

You want to see something like:

```
status=sent
```

The key idea:

Nagios

↓

mail command / notification command



Postfix



Amazon SES



Gmail

Don't troubleshoot all four layers simultaneously. First prove Postfix
→ SES works.

PART B — Now connect Nagios to email

Your Nagios configuration has to answer three questions:

WHO receives the email?



contacts.cfg

WHEN should they receive it?



host/service notification settings

HOW should Nagios send it?



commands.cfg

1. contacts.cfg — WHO?

Open:

```
sudo nano /usr/local/nagios/etc/objects/contacts.cfg
```

You'll find a contact such as:

```
define contact {  
    contact_name      nagiosadmin  
    use               generic-contact  
    alias             Nagios Admin  
    email             YOUR_VERIFIED_EMAIL  
}
```

The important line is:

```
email YOUR_VERIFIED_EMAIL
```

This tells Nagios:

This is the email address belonging to this contact.

But here's the important part

Adding:

```
email YOUR_VERIFIED_EMAIL
```

does NOT automatically mean Nagios will send email.

The contact also needs notification commands.

For example:

```
define contact {  
    contact_name      nagiosadmin  
    use               generic-contact  
    alias             Nagios Admin  
    email             YOUR_VERIFIED_EMAIL  
    host_notification_commands    notify-host-by-email
```

```
    service_notification_commands    notify-service-by-email
}
```

Now we're saying:

```
nagiosadmin
```

↓

```
email = YOUR_VERIFIED_EMAIL
```

Host notification

↓

```
notify-host-by-email
```

Service notification

↓

```
notify-service-by-email
```

2. commands.cfg — Email Notification Commands

Edit:

```
sudo nano /usr/local/nagios/etc/objects/commands.cfg
```

Define the email commands for host and service notifications:

```
define command {
    command_name    notify-host-by-email

    command_line    /usr/bin/printf "%b" "***** Nagios
*****\n\nNotification Type: $NOTIFICATIONTYPE$\nHost:
$HOSTNAME$\nState: $HOSTSTATE$\nAddress:
$HOSTADDRESS$\nInfo: $HOSTOUTPUT$\n\nDate/Time:
$LONGDATETIME$\n" | /usr/bin/mail -r
```

```
"YOUR_VERIFIED_SES_EMAIL" -s "*** $NOTIFICATIONTYPE$
Host Alert: $HOSTNAME$ is $HOSTSTATE$ ***"
$CONTACTEMAIL$
}
```

```
define command {
    command_name    notify-service-by-email
    command_line    /usr/bin/printf "%b" "***** Nagios
*****\n\nNotification Type: $NOTIFICATIONTYPE$\n\nService:
$SERVICEDESC$\nHost: $HOSTALIAS$\nAddress:
$HOSTADDRESS$\nState: $SERVICESTATE$\n\nInfo:
$SERVICEOUTPUT$\n\nDate/Time: $LONGDATETIME$\n" |
/usr/bin/mail -r "YOUR_VERIFIED_SES_EMAIL" -s "***
$NOTIFICATIONTYPE$ Service Alert:
$HOSTALIAS$/$SERVICEDESC$ is $SERVICESTATE$ ***"
$CONTACTEMAIL$
}
```

Replace YOUR_VERIFIED_SES_EMAIL with an email address
verified in Amazon SES.

- printf → creates the email body.
- -r → sets the **From** address.
- -s → sets the subject.
- \$CONTACTEMAIL\$ → sets the **To** address.

Validate after editing:

```
sudo nagios -v /usr/local/nagios/etc/nagios.cfg
```

```
sudo systemctl restart nagios
```



```
ubuntu@nagios-mastery: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_service.py  
#!/usr/bin/env python3  
  
import sys  
import subprocess  
  
SERVICE = sys.argv[1] if len(sys.argv) > 1 else "nginx"  
  
try:  
    result = subprocess.run(  
        ["systemctl", "is-active", SERVICE],  
        capture_output=True,  
        text=True  
    )  
except Exception:  
    print(f"UNKNOWN unable to check {SERVICE}")  
    sys.exit(3)  
  
state = result.stdout.strip()  
  
if result.returncode == 0 and state == "active":  
    [ Read 35 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace  ^U Paste    ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-mastery: ~  
GNU nano 6.2 /usr/local/nagios/libexec/check_service.py  
  
state = result.stdout.strip()  
  
if result.returncode == 0 and state == "active":  
    print(f"Service is OK: {SERVICE}")  
    sys.exit(0)  
  
elif state == "inactive" or state == "failed":  
    print(f"CRITICAL {SERVICE} is not running")  
    sys.exit(2)  
elif state == "activating" or state == "deactivating":  
    print(f"CRITICAL {SERVICE} is in {state} state")  
    sys.exit(2)  
elif state == "unknown" or state == "not-found":  
    print(f"UNKNOWN {SERVICE} does not exist")  
    sys.exit(3)  
else:  
    print(f"UNKNOWN unable to decide state of {SERVICE}")  
    sys.exit(3)  
|  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace  ^U Paste    ^J Justify  ^_ Go To Line
```

```
ubuntu@nagios-mastery: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg
define command {
    command_name    check_local_disk
    command_line    $USER1$/check_disk -w $ARG1$ -c $ARG2$ -p $ARG3$
}

define command {
    command_name    check_disk_usage
    command_line    $USER1$/check_disk_usage $ARG1$ $ARG2$ $ARG3$
}

define command {
    command_name    check_service
    command_line    $USER1$/check_service.py $ARG1$
}

define command {
|
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mastery: ~
GNU nano 6.2 /usr/local/nagios/etc/objects/localhost.cfg
define service {
    use local-service
    host_name localhost
    service_description Nginx System Service check
    check_command check_service!nginx
    check_interval 1
    retry_interval 1
    max_check_attempts 3

    check_period 24x7

    notifications_enabled 1
    notification_interval 30
    notification_period 24x7
    notification_options w,c,u,r

    contact_groups admins
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

13.201.137.246/nagios/

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer

Nagios

General

- Home
- Documentation
- Current Status
- Tactical Overview
- Map
- Hosts
- Services
- Host Groups
- Summary
- Grid
- Service Groups
- Summary
- Grid
- Problems
- Services (Unhandled)
- Hosts (Unhandled)
- Network Outages
- Quick Search:

Reports

- Availability
- Trends
- Alerts
- History
- Summary
- Histogram
- Notifications
- Event Log
- System
- Comments
- Downtime
- Process Info
- Performance Info

Service Information

Last Updated: Sat Sep 5 11:38:51 UTC 2026
Updated every 90 seconds
Nagios® Core™ 4.5.10 - www.nagios.org
Logged in as nagiosadmin

View Information For This Host
View Status Detail For This Host
View Alert History For This Service
View Trends For This Service
View Alert Histogram For This Service
View Availability Report For This Service
View Notifications For This Service

Service

Nginx System Service check

On Host
localhost
(localhost)

Member of
No servicegroups.

127.0.0.1

Service State Information

Current Status: CRITICAL (for 0d 0h 0m 20s)
Status Information: CRITICAL nginx is not running
Performance Data:
Current Attempt: 3/3 (HARD state)
Last Check Time: 09-05-2026 11:38:31
Check Type: ACTIVE
Check Latency / Duration: 0.001 / 0.030 seconds
Next Scheduled Check: 09-05-2026 11:39:31
Last State Change: 09-05-2026 11:38:31
Last Notification: 09-05-2026 11:38:31 (notification f)
Is This Service Flapping? NO (6.25% state change)
In Scheduled Downtime? NO
Last Update: 09-05-2026 11:38:45 (0d 0h 0m 6s ago)

Active Checks: ENABLED
Passive Checks: ENABLED
Obsessing: ENABLED
Notifications: ENABLED
Event Handler: ENABLED
Flap Detection: ENABLED

Service Commands

- ✗ Disable active checks of this service
- 🕒 Re-schedule the next check of this service
- ❓ Submit passive check result for this service
- ✗ Stop accepting passive checks for this service
- ✓ Stop obsessing over this service
- ✓ Acknowledge this service problem
- ✗ Disable notifications for this service
- 🕒 Delay next service notification
- 📧 Send custom service notification
- 🕒 Schedule downtime for this service
- ✗ Disable event handler for this service
- ✗ Disable flap detection for this service
- ✗ Clear flapping state for this service

Service Comments

🗨 Add a new comment 🗑 Delete all comments

Entry Time	Author	Comment	Comment ID	Persistent	Type	Expires	Actions
This service has no comments associated with it							

Page Four

mail.google.com/mail/u/0/#spam/FMfcgzQHWLLZxTjD8kNLxkQXPhQfnGb

Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Progra... CVB in Unsupervise... Red Hat Academy System Design Primer

Gmail

in:spam

Upgrade

2 of 69

Delete forever Not spam

**** PROBLEM Service Alert: localhost/Nginx System Service check is CRITICAL **** Spam x

sreevishnu0407@gmail.com via amazonse.com

17:08 (6 minutes ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report as not spam

**** Nagios ****

Notification Type: PROBLEM

Service: Nginx System Service check
Host: localhost
Address: 127.0.0.1
State: CRITICAL

Date/Time: Sat Sep 5 11:38:31 UTC 2026

Additional Info:

CRITICAL nginx is not running

6. Custom Bash Load Monitoring Plugin with getopts, Threshold-Based Alerting and Nagios \$ARG\$ Integration

```
ubuntu@nagios-mastery: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_load_simple *
#!/bin/bash

while getopts "w:c:" opt; do
  case "$opt" in
    w)
      WARNING=$OPTARG
      ;;
    c)
      CRITICAL=$OPTARG
      ;;
    *)
      echo "UNKNOWN INVALID ARGUMENTS"
      exit 3
      ;;
  esac
done

if [ -z "$WARNING" ] || [ -z "$CRITICAL" ]; then
  echo "UNKNOWN - Missing warning or critical threshold"
  exit 3
fi
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

```
ubuntu@nagios-mastery: ~
GNU nano 6.2 /usr/local/nagios/libexec/check_load_simple *

if [ -z "$WARNING" ] || [ -z "$CRITICAL" ]; then
  echo "UNKNOWN - Missing warning or critical threshold"
  exit 3
fi

LOAD=$(awk '{print $1}' /proc/loadavg)

if awk "BEGIN {exit !($LOAD>=$CRITICAL)}"; then
  echo "CRITICAL - Load average is $LOAD"
  exit 2
elif awk "BEGIN {exit !($LOAD>=$WARNING)}"; then
  echo "WARNING - Load average is $LOAD"
  exit 1
else
  echo "OK! Load Average is $LOAD"
  exit 0
fi
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

```
ubuntu@nagios-mastery: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/commands.cfg  
  
command_name check_service  
command_line $USER1$/check_service.py $ARG1$  
}  
  
define command {  
    command_name check_load_simple  
    command_line $USER1$/check_load_simple $ARG1$ $ARG2$ $ARG3$ $ARG4$  
}  
  
define command {  
    command_name check_local_load  
    command_line $USER1$/check_load -w $ARG1$ -c $ARG2$  
}  
  
define command {  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
ubuntu@nagios-mastery: ~  
GNU nano 6.2 /usr/local/nagios/etc/objects/localhost.cfg  
  
define service {  
    use local-service  
    host_name localhost  
    service_description System Load Average  
  
    check_command check_load_simple!-w!0.01!-c!0.02  
  
    check_interval 1  
    retry_interval 1  
    max_check_attempts 3  
  
    notifications_enabled 1  
    notification_interval 30  
    notification_period 24x7  
    notification_options w,u,c,r  
  
    contact_groups admins  
}  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Dance Tour

Page 104

13.201.137.246/nagios/

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Reports

Availability

Trends

Alerts

History

Summary

Histogram

Notifications

Event Log

System

Comments

Downtime

Process Info

Performance Info

Service Information

Last updated: Sat Sep 5 12:29:15 UTC 2026

Updated every 90 seconds

Nagios® Core™ 4.5.10 - www.nagios.org

Logged in as nagiosadmin

View Information For This Host

View Status Detail For This Host

View Alert History For This Service

View Trends For This Service

View Alert Histogram For This Service

View Availability Report For This Service

View Notifications For This Service

Service

System Load Average

On Host

localhost

(localhost)

Member of

No servicegroups.

127.0.0.1

Service State Information

Current Status: **CRITICAL** (for 0d 0h 0m 5s)

Status Information: CRITICAL - Load average is 1.02

Performance Data:

Current Attempt: 3/3 (HARD state)

Last Check Time: 09-05-2026 12:29:10

Check Type: ACTIVE

Check Latency / Duration: 0.001 / 0.013 seconds

Next Scheduled Check: 09-05-2026 12:30:10

Last State Change: 09-05-2026 12:29:10

Last Notification: 09-05-2026 12:29:10 (notification 1)

Is This Service Flapping? **NO** (6.25% state change)

In Scheduled Downtime? **NO**

Last Update: 09-05-2026 12:29:11 (0d 0h 0m 4s ago)

Active Checks: **ENABLED**

Passive Checks: **ENABLED**

Obsessing: **ENABLED**

Notifications: **ENABLED**

Event Handler: **ENABLED**

Flap Detection: **ENABLED**

Service Commands

Disable active checks of this service

Re-schedule the next check of this service

Submit passive check result for this service

Stop accepting passive checks for this service

Stop obsessing over this service

Acknowledge this service problem

Disable notifications for this service

Delay next service notification

Send custom service notification

Schedule downtime for this service

Disable event handler for this service

Disable flap detection for this service

Clear flapping state for this service

Service Comments

Add a new comment

Delete all comments

Entry Time Author Comment Comment ID Persistent Type Expires Actions

This service has no comments associated with it

mail.google.com/mail/u/0/#spam/FMfcgzQHWLLZzGnVwXMFgKPygPCNtlh

Gmail

in:spam

Delete forever Not spam

1 of 70

**** PROBLEM Service Alert: localhost/System Load Average is CRITICAL **** Spam x

sreevishnu0407@gmail.com [via](#) amazonses.com

to me

17:59 (0 minutes ago)

Why is this message in spam? This message is similar to messages that were identified as spam in the past.

Report as not spam

***** Nagios *****

Notification Type: PROBLEM

Service: System Load Average

Host: localhost

Address: 127.0.0.1

State: CRITICAL

Date/Time: Sat Sep 5 12:29:10 UTC 2026

Additional Info:

CRITICAL - Load average is 1.02

```
ubuntu@nagios-mastery: ~  
ubuntu@nagios-mastery:~$ kill yes  
[1]+  Terminated                  yes > /dev/null  
ubuntu@nagios-mastery:~$ jobs  
ubuntu@nagios-mastery:~$ cat /proc/loadavg  
0.99 0.67 0.30 1/160 3756
```


mail.google.com/mail/u/0/#spam/FMfcgZQHLLZxbMpzBrTCIbMcdFrmcZC

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer

Machine LearningNeural Networks ar...Density-based clust...Fisher LDA-StanfordCompetitive Progra...CVB in Unsupervise...Red Hat AcademySystem Design Primer

Gmail

in:spam

Upgrade

1 of 73

18:06 (28 minutes ago)

**** Nagios ****

Notification Type: RECOVERY

Service: System Load Average

Host: localhost

Address: 127.0.0.1

State: OK

Date/Time: Sat Sep 5 12:36:09 UTC 2026

Additional Info:

OK! Load Average is 0.00

**** RECOVERY Service Alert: localhost/System Load Average is OK ****

sreevishnu0407@gmail.com

18:06 (28 minutes ago)

Report as not spam